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NITTE (Deemed to be University)

(Established under Sec.(3) of UGC Act, 1956)

First Semester M.Tech (CBCS) Degree Examinations, December 2025

25MCT132 – AUTOTRONICS AND VEHICLE INTELLIGENCE

(For MCT)

Duration: 3 Hours]

[Max. Marks: 100

Note: Answer **Five full** questions choosing **Two full** questions from **Module – I & Module II** & **Two full** questions from **Module – III & Module IV** each and **One full** question from **Module – V**.

Module – I & Module - II		Marks	BT*	CO*	PO*
1. a)	Explain the requirements of a fuel supply system with a neat schematic.	10	2	1	1
b)	Explain how principles of the ignition system contribute to improved engine performance through electronic spark timing control.	10	3	2	2
2. a)	Describe the working of a throttle-body injection system with a diagram.	10	2	1	2
b)	Explain how crash-sensing mechanisms enable the functioning of multi-stage airbag deployment systems.	10	3	2	1
3. a)	Explain the construction and operation of a magneto ignition system.	10	2	1	1
b)	Using illustrations, demonstrate how temperature and pressure sensors are integrated in automotive diagnostic systems.	10	3	2	2
Module – III & Module - IV					
4. a)	Explain catalytic converter types with neat sketches.	10	2	3	1
b)	Explain how adaptive control concepts are utilized in the functioning of Adaptive Cruise Control.	10	3	4	1
5. a)	Describe the components of hybrid vehicle architecture with diagrams.	10	2	3	2
b)	Using a block diagram, demonstrate how drowsy-driver detection systems evaluate eye and motion parameters.	10	3	4	2
6. a)	Explain the role of emission standards in engine performance regulation.	10	2	3	1
b)	Explain how lane-keeping assist systems use crash-avoidance principles to reduce driver workload.	10	3	4	1
Module – V					
7. a)	Explain the working of a MEMS accelerometer used in automotive systems.	10	2	5	1
b)	Using a vision-processing diagram, illustrate how autonomous vehicles detect and interpret lane markings.	10	3	5	2
8. a)	Describe the architecture of a basic intelligent transportation system (ITS).	10	2	5	2
b)	Explain how autonomous vehicles make navigation decisions using multi-sensor fusion techniques.	10	3	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
